

Unifying Scene-Object Congruency and Incongruency Benefits in Object Perception

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While the influence of scene-object semantic congruency on object perception is well established, the direction of the influence remains controversial. We address this issue by presenting an innovative approach that uses a vector-space semantic model to quantify scene-object congruency as a continuous variable. By exploring a wide range of congruency values and using multiple experimental tasks, we aimed to comprehensively investigate the relationship between scene-object congruency and object perception across four experiments ($N = 543$). We found a robust positive U-shaped relationship between scene-object congruency and accuracy in an exemplar identification task (Experiment 1), which persisted across different stimulus sets (reanalysis of the previous data set) and scene presentation times (Experiment 2). To explore the mechanisms underlying the U-shaped relationship, we propose that scene-object congruency affects object perception in two ways: by influencing attention and thus processing amount and by directly influencing processing efficiency. In Experiment 3, a change detection task emphasizing attentional processes revealed an inverted U-shaped relationship between congruency and reaction times, suggesting that both congruent and incongruent contexts improve object perception by increasing attention. Experiment 4 introduced a cue to neutralize congruency-based attention, focusing on processing efficiency in an identification task. Notably, only a congruency benefit emerged, while an incongruency benefit was absent, suggesting that congruency uniquely facilitates processing efficiency to improve object perception. In summary, our study provides compelling evidence for the coexistence of congruency and incongruency benefits, reconciling previous contradictions and providing a unifying framework for understanding the relationship between scene-object congruency and object perception.

Public Significance Statement

The present study introduces an innovative method that employs a vector-space semantic model to quantify scene-object congruency, providing a fresh perspective on the long-standing debate surrounding the benefits of congruency and incongruency. Through comprehensive investigations of stimuli across the congruency continuum, our findings reveal a robust U-shaped relationship between scene-object congruency and object perception. We propose that scene-object congruency influences object perception in two distinct ways. First, both congruent and incongruent scene contexts enhance object perception by increasing attention and, consequently, the amount of processing. Second, congruency uniquely improves processing efficiency, which further contributes to enhanced object perception.

Keywords: scene-object congruency, perception, context, attention, preregistered

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Humans are remarkably efficient at perceiving objects embedded in natural scenes. This efficiency is supported by semantic associations between objects and scenes, with objects exhibiting strong semantic congruency with the scene in which they are typically embedded (e.g., Peelen et al., 2024). For example, compared to a watermelon, a football is more likely to appear in a football field, where scene-object congruency is stronger. Extensive research has focused on the role of scene-object congruency in object perception. However, a unified framework has yet to be reached.

Congruency Benefit

Extensive research indicates that semantic congruency between scenes and objects enhances object perception, termed the “congruency benefit” (for reviews, see Bar, 2004; Oliva & Torralba, 2007). This phenomenon aligns well with predictive processing theory, often elucidated through a Bayesian framework. According to this theory, our perceptual representations are biased toward prior expectations (Clark, 2013; de Lange et al., 2018; Friston, 2005; Press et al., 2020; Rao & Ballard, 1999). Scene contexts can generate expectations about encountered objects, facilitating their integration with prior knowledge and increasing *processing efficiency*. Consequently, objects congruent with the scene are identified more easily and quickly.

This theoretical explanation is empirically supported by behavioral studies, which have consistently shown that objects are named and categorized more quickly and accurately in congruent scene contexts than in incongruent ones, especially when the stimuli are presented for a brief duration (e.g., less than 150 ms; Boyce et al., 1989; Davenport, 2007; Davenport & Potter, 2004; Fize et al., 2011; Munneke et al., 2013; Palmer, 1975). This congruency benefit extends beyond semantic tasks, persisting in nonsemantic scenarios. For instance, Chen et al. (2022) found that when a target object was presented for only 83 ms at the cued location within a scene, the identification of the target object from its exemplars still showed a congruency benefit. Similarly, in a study by Rossel et al. (2023), when participants were tasked with matching blurred images based on perceived blur, scene congruent objects were perceived as less blurred compared to incongruent ones, highlighting the role of scenes in sharpening object perception under ambiguity.

Consistent with the behavioral findings, Krugliak et al. (2024) examined the relationship between electroencephalogram patterns during scene perception and a semantic model based on property norms. Their results indicated a shorter duration of semantic integration for congruent objects compared to incongruent objects, suggesting more efficient semantic processing for objects in congruent scenes. In another electroencephalogram study, object representations were better decoded for objects in congruent scenes when the object appeared briefly at the cued location within the scene, providing neural evidence for the congruency benefit (Chen et al., 2022).

Incongruency Benefit

Despite the prevalence of congruency benefits, a substantial body of research suggests a perceptual advantage for objects that are incongruent with the scene (e.g., a watermelon on a football field), termed the “incongruency benefit” (or “congruency cost”; e.g., Hollingworth & Henderson, 2000; Spaak et al., 2022). Also rooted in predictive processing theory, it has been proposed that incongruent objects that defy expectations lead to a prediction error.

Resolving this error requires that observers engage in more active sampling (devote more attention) to these incongruent objects, thereby increasing the amount of processing (Spaak et al., 2022). Assuming that increased processing increases subjective awareness (Anzulewicz et al., 2015; Windey et al., 2013), perceptual performance for incongruent objects would be improved relative to congruent objects.

Building on this theoretical foundation, empirical evidence provides robust support for the incongruency benefit phenomenon. Studies have found that changes in incongruent objects are detected faster than in congruent ones (Hollingworth & Henderson, 2000; LaPointe et al., 2013; Mack et al., 2017; Ortiz-Tudela et al., 2023; Spaak et al., 2022). Furthermore, eye-tracking research consistently indicates a preference for fixating on incongruent objects—earlier, longer, and more frequently—than on congruent ones (Bonitz & Gordon, 2008; Cornelissen & Vö, 2017; Vö & Henderson, 2009, 2011). This effect persists even in purely perceptual tasks, such as the exemplar identification task used by Spaak et al. (2022), where incongruent conditions yielded superior perceptual performance, leading to the conclusion that scene contexts impair the perception of congruent objects.

Coexisting Congruency and Incongruency Benefits

The above studies have produced seemingly contradictory results. How can we explain the contradiction between the scene-object congruency and incongruency benefits?

We propose that the contradictions in scene-object congruency research stem from its conventional binary classification—categorizing stimuli as congruent or incongruent based on subjective ratings. This approach results in labeling improved performance in congruent conditions as “congruency benefit” and superior performance in incongruent conditions as “incongruency benefit” or “congruency cost.” However, it overlooks the possibility that the perception of scene congruent and incongruent objects may be differentially enhanced relative to those of intermediate congruency. To fill this gap, we argue that scene-object congruency should be treated as a continuum, allowing for a more nuanced investigation of its effects on object perception.

Our view of congruency as a continuous variable has received some theoretical support. Oliva and Torralba (2007) introduced the concept of scene-object relationship strength, which they proposed depends on the degree of binding between scenes and objects. For example, a cup may appear in scenes such as a kitchen or classroom and be moderately associated with them, while a bowling ball may be strongly associated with a bowling alley but weakly associated with a kitchen. Oliva and Torralba (2007) proposed that the stronger the scene-object relationship, the greater the influence of the scene on the object. In addition, the opposing process theory by Press et al. (2020) provides a framework for understanding how congruency and incongruency benefits manifest in two stages. Initially, perception relies on prior knowledge, which facilitates the rapid generation of accurate perceptual experiences—corresponding to congruency benefits. However, when a highly unexpected event occurs, a second process is triggered to emphasize it—corresponding to incongruency benefits. A key aspect of this theory is that this latter process is activated specifically when an event is sufficiently “surprising.” Thus, the emergence of incongruency benefits is contingent on the degree of incongruency.

Empirical evidence further underscores the need to treat scene-object congruency as a continuous variable. Studies using change detection tasks have consistently found incongruency benefits (e.g., Hollingworth & Henderson, 2000; LaPointe et al., 2013; Mack et al., 2017; Spaak et al., 2022). However, in contexts where objects are crucial for scene comprehension—highly scene congruent situations—the effect shifts to a congruency benefit (Rensink et al., 1997). Eye-tracking research supports this continuum approach by showing that when congruency is divided into two levels, early fixations were on the incongruent objects, suggesting an incongruency benefit (Bonitz & Gordon, 2008). However, a more recent and pioneering study by Damiano et al. (2024) offers a deeper understanding by treating congruency as a continuous variable. Their findings reveal that both highly congruent and highly incongruent objects attract more fixations than moderately congruent ones, underscoring a U-shaped pattern. This groundbreaking research emphasizes that the effects of congruency on object perception are influenced by its position on a continuum, highlighting the potential for confounding results when selecting stimuli without considering the full spectrum of congruency. It leads our study to advocate for a continuous variable treatment of scene-object congruency, aiming for a comprehensive exploration of its effects on perception.

Another source of contradictions in previous research stems from the distinct mechanisms driving the benefits of congruency and incongruency. We suggest that scene-object congruency differentially impacts both attention (thereby influencing the amount of processing) and processing efficiency, collectively shaping object perception in diverse manners. This perspective is evidenced in empirical studies utilizing various experimental paradigms.

Specifically, attention-demanding tasks, such as change detection (e.g., Hollingworth & Henderson, 2000) and exemplar identification without attentional cues (e.g., Spaak et al., 2022), typically reveal incongruency benefits. However, as previously reviewed, the relationship between congruency and attention may follow a U-shaped pattern (Damiano et al., 2024), with both highly congruent and highly incongruent objects attracting more attention than moderately congruent ones, thereby influencing object perception. Conversely, tasks that focus on exemplar identification processes, such as categorization (Fize et al., 2011), naming (Davenport & Potter, 2004), and exemplar identification with explicit attentional cues (Chen et al., 2022), predominantly demonstrate a congruency benefit. These tasks are characterized by short stimulus presentation times, highlighting that congruent objects show higher processing efficiency than incongruent objects within a limited processing time. This evidence collectively suggests that the impact of scene-object congruency on object perception is multifaceted and requires comprehensive investigation.

The Present Study

This study aimed to comprehensively explore the relationship between scene-object congruency and object perception. In addressing the challenges posed by the binary conceptualization of scene-object congruency, our study introduces a novel approach. We leverage a vector-space semantic model to provide a more objective and quantifiable measure of scene-object congruency. This method utilizes semantic vectors from ConceptNet Numberbatch, which integrates word usage in written text with crowdsourced foundational knowledge about the world (Günther et al., 2019), and has previously

been applied to scene perception (Damiano et al., 2024; Hayes & Henderson, 2021). Such an approach allows us to explore the continuum of scene-object congruency precisely, facilitating a nuanced exploration of the relationship between scene-object congruency and object perception.

To further explore the second source of contradictions—specifically, the different mechanisms behind congruency and incongruency benefits—our study implemented three experimental paradigms. We began with the purely perceptual, bias-free exemplar identification task used by Spaak et al. (2022), asking participants to freely view a scene and make a two-alternative forced choice (2AFC) to identify the previously presented object in the scene (Experiments 1 and 2). Considering that scene-object congruency's influence on perception may arise from its effects on attention (subsequently affecting processing amounts) and processing efficiency, we furthered our exploration using two additional experimental paradigms, each designed to focus on one of these key aspects. In Experiment 3, we utilized a change detection task, which underscores the role of attention capture (LaPointe et al., 2013) by requiring participants to detect and localize the changing object. Experiment 4 added an attentional cue to the exemplar identification task, neutralizing attentional benefits to isolate processing efficiency advantages. This approach provides a nuanced examination of the impact of scene-object congruency on object perception, distinguishing between the effects of processing amount and efficiency.

The literature reviewed above suggests a potential positive linear relationship between scene-object congruency and processing efficiency (e.g., Chen et al., 2022), along with a U-shaped relationship between scene-object congruency and attention (e.g., Bonitz & Gordon, 2008; Damiano et al., 2024; Hayes & Henderson, 2021). Therefore, experimental paradigms focusing on different aspects would be associated with different hypotheses (see Figure 1). For the exemplar identification paradigm (Experiments 1 and 2), which assesses both attention and processing efficiency, we predict a positive U-shaped relationship between scene-object congruency and identification performance. In the change detection paradigm (Experiment 3), which is primarily influenced by attention processes, we hypothesize an inverted U-shaped relationship between scene-object congruency and reaction times (RTs). In the cue-present identification paradigm (Experiment 4), where attention advantages are mitigated by explicit cues, we hypothesize a positive linear relationship between scene-object congruency and identification performance.

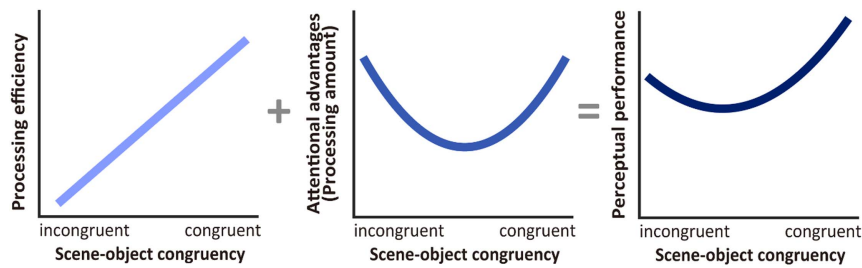
Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. Data were analyzed using R (R Core Team, 2018) and JASP (JASP Team, 2020). The data and code of this article are available on the project's Open Science Framework (OSF) page at <https://osf.io/m3yej>. Three experiments in this article were preregistered: Experiment 1b (<https://osf.io/q3cn5>), Experiment 3 (<https://osf.io/ng9ec>), and the control experiment (see Supplemental Material 1, <https://osf.io/wm3kb>).

Experiment 1a: Exemplar Identification Task

The purpose of Experiment 1 is to examine the relationship between scene-object congruency and identification performance.

Figure 1
Experimental Hypothesis



Note. These three panels illustrate our predictions regarding the relationship between scene-object congruency and object perception. We posit a positive U-shaped relationship between scene-object congruency and perceptual performance. This positive U-shaped pattern emerges from the linear relationship between congruency and processing efficiency, overlaid by a U-shaped relationship between congruency and attention (processing amount). See the online article for the color version of this figure.

We used an exemplar identification task similar to Experiment 2 in Spaak et al. (2022), a paradigm recognized for its bias-free and purely perceptual nature. Participants were shown a scene and then were asked to identify the exemplar they had previously seen in a 2AFC task. To examine the relationship comprehensively, we used ConceptNet Numberbatch to quantify scene-object congruency of our stimuli. Our stimuli spanned a wide range of congruency levels, from very low (e.g., bathroom and umbrella, with a congruency score of 0.06) to very high (e.g., bathroom and towel, with a congruency score of 0.86). We hypothesize a positive U-shaped relationship between scene-object congruency and performance in the identification task.

Method

Participants

A total of 69 participants (37 participants reported their gender as female, 32 as male; age: $M = 23.01$ years, $SD = 2.55$) completed the experiment online through the Naodao platform (<https://www.naodao.com/>) and passed the catch trials (see Procedure for details). All participants provided signed informed consent, had normal or corrected-to-normal vision, and were paid 5 Chinese yuan each. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

The sample size of Experiment 1a was determined for convenience. The observed power computed by the package *SIMR* (Version 1.0.7; Green & MacLeod, 2016) was 100.00% (95% CI [96.38%, 100.00%]).

Stimuli

Scene Stimuli. We aimed to select scene stimuli with a broad range of scene-object congruencies. To achieve this, we chose not to use a perfectly controlled, matched stimulus set. This was because objects that are highly congruent with the scene (e.g., a closestool that is highly congruent with a bathroom) are often rare in other scenes, making it difficult to find matches for moderately congruent scenes and limiting the range of stimuli available.

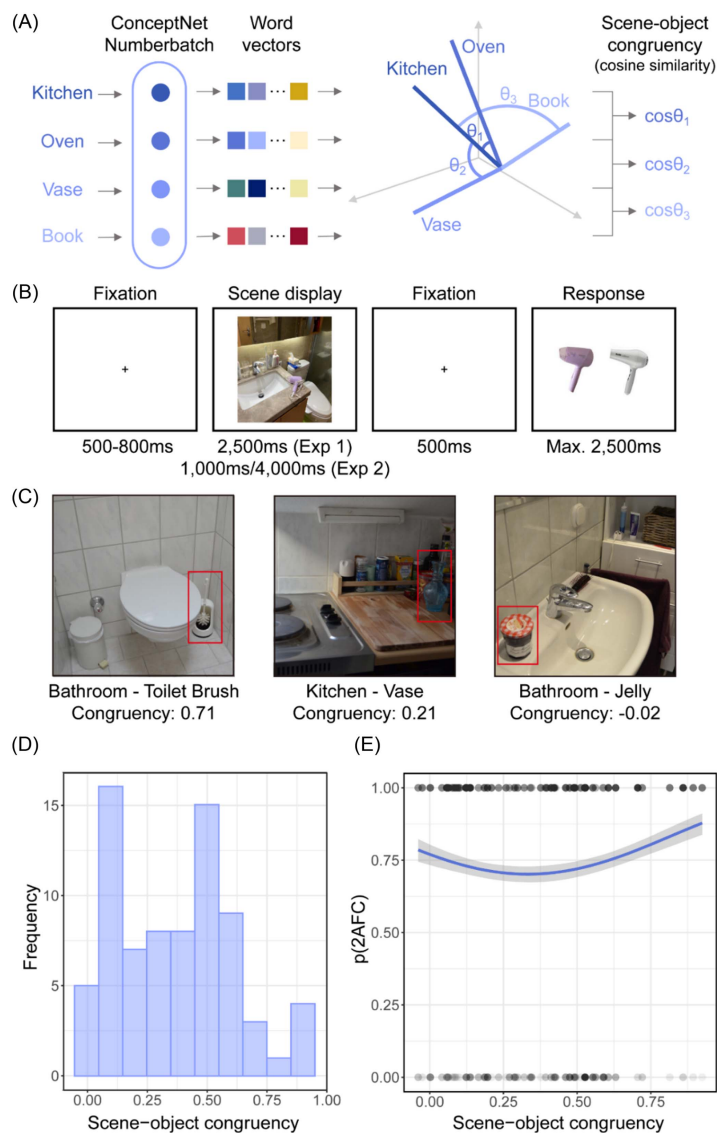
Our stimuli were sourced from the SCEGRAM database (Öhlschlager & Vö, 2017), the Berlin Object in Scene database (Mohr et al., 2016), and the Internet. However, due to potential cultural differences between the backgrounds represented in these databases and those of our study participants, some of the objects represented in these databases may be relatively unfamiliar to Chinese participants. To address this concern, all selected stimuli underwent a rigorous screening process using a questionnaire. An additional 12 participants (seven females; $M = 22.83$ years, $SD = 1.07$ years; undergraduate and graduate students of Sun Yat-sen University) completed the questionnaire. They were asked to name the scene and the target objects and to rate the congruency between the two using a 7-point Likert scale. The naming results were then cross-checked by two independent observers. Only stimuli correctly identified by more than 10 participants were retained for inclusion in the stimulus set. Ultimately, 90 scene images were included in the stimulus set, 49 of which were obtained from the SCEGRAM database (Öhlschlager & Vö, 2017), 11 from the Berlin Object in Scene database (Mohr et al., 2016), and an additional 30 from the Internet.

We used the Chinese version of ConceptNet Numberbatch to generate semantic vectors. ConceptNet Numberbatch is a vector-space semantic model that combines the way words are used in text with basic knowledge about the world (Günther et al., 2019). We calculated the semantic similarity between the scene vector and the object vector (i.e., the cosine similarity of the vectors; see Figure 2A). The calculated semantic similarity ranges from -0.04 to 0.92 , that is, including both highly incongruent and highly congruent cases. The computed semantic similarity was found to be significantly correlated with the subjective ratings in the preexperiment questionnaire ($r = .61$, $p < .001$), indicating that it is reasonable to use this model similarity as a measure of scene-object congruency.

Alternatives. In the two alternatives of each trial, there was always a target object appearing in the scene and a lure item. The lure items were obtained from the Internet. Lures always belonged to the same category as the target, but were different exemplars.

To mitigate potential confounds in trial difficulty caused by the similarity between alternatives, we implemented two approaches. First, we calculated the image similarity between targets and lures during lure selection. The similarity between two alternatives was

Figure 2
Design and Results for Experiment 1a



Note. (A) illustrates how scene-object congruency is quantified. Using the example of the scene “kitchen” and the objects “oven,” “vase,” and “book,” we obtained vectors for each word in a high-dimensional space from ConceptNet Numberbatch (here, only three dimensions are visualized for clarity; in reality, the word vectors are 300-dimensional). We then computed the cosine similarity between the scene and object word vectors. The experimental procedure of Experiments 1 and 2 is shown in (B). After a 500–800 ms fixation, a scene display was presented for 2,500 ms in Experiment 1 and 1,000 ms and 4,000 ms in Experiment 2. Participants were asked to make a two-alternative forced choice (2AFC) to identify which object in the scene had been presented previously. (C) shows examples of scene-object pairings used in our study, along with their congruency scores. The target object is highlighted in the red box. The frequency distribution of scene-object congruency for stimuli used in Experiments 1 and 2 is shown in (D). From *SCEGRAM Database*, by the Scene Grammar Lab, Goethe University Frankfurt (<https://www.scenegrammarlab.com/databases/scegram-database/>). (E) shows the 2AFC results for Experiment 1a. The values on the x -axis in (E) were the scene-object congruency computed from the scene-object labels. The blue line represents the predicted second-order polynomial relationship, and the shaded area around the blue line indicates the 95% credible intervals of the prediction. Scatter points represent the density of individual trials in which the 2AFC was correct (1) or incorrect (0). Max = maximum; Exp = experiment. See the online article for the color version of this figure.

quantified using AlexNet, a well-trained convolutional neural network for image classification (Krizhevsky et al., 2012). We controlled for alternatives similarity to be between 0.50 and 0.80.

Second, we conducted an additional control experiment (pre-registered on the OSF at <https://osf.io/wm3kb>). In this control experiment, we extracted the target from the scene and placed it on a white background using Adobe Photoshop, essentially removing the scene information from each stimulus. The control experiment procedure was consistent with Experiment 1a, except that stimuli were the scene-free versions of those used in Experiment 1a. Participants were required to view the stimuli for 2,500 ms and then make a binary forced choice after a blank screen to select the target they had just viewed. Stimuli with low accuracy in this control experiment might suffer from poor alternative discriminability and overly challenging trial difficulty. Therefore, we excluded stimuli from the further analysis if their accuracy in the control experiment was less than 1 *SD* below the mean accuracy. We applied a relatively conservative criterion to ensure that the observed U-shaped relationship was not driven by outliers. It is worth noting that if no stimuli were excluded, the U-shaped effect would actually become more pronounced (see [Supplemental Material 1](#) for details on the results of the control experiment and Experiment 1a without excluding stimuli from the analysis). After the screening process, 76 stimuli remained. The semantic similarity scores for the remaining stimuli varied between -0.04 and 0.92 , still including cases of both high incongruency and high congruency (see [Figure 2C](#) and [2D](#)).

Confounding Variables. We also considered some confounding variables: saliency, size, center proximity, complexity, and familiarity, which were included in the further regression model analysis to control for their potential effects. For saliency, we calculated the saliency rank within 15 simulated fixations using the Saliency Toolbox (Walther & Koch, 2006). It has a value of 1 for the most salient area of an image. If none of the simulated fixations fell within the area of the target object, the fixation rank was set to 15. Size refers to how big the target object is. Center proximity refers to the distance between the target object and the center of the scene. Complexity refers to the number of unique object categories (Chai et al., 2010). For familiarity, an additional 34 participants (23 females; $M = 21.70$ years, $SD = 1.54$ years; undergraduate and graduate students of Sun Yat-sen University) provided ratings on a 10-point Likert scale for the familiarity of each item word. The mean ratings of all participants were then included as a covariate in subsequent analyses.

Procedure

This experiment was depicted in [Figure 2B](#). Each trial started with a fixation cross (800–1,000 ms, randomly selected from a uniform distribution). Subsequently, a scene display lasting 2,500 ms is presented on the screen. After another fixation cross (500 ms), two alternatives were presented on the screen. Participants performed a 2AFC task in which they selected the alternative that had appeared in the scene. Participants pressed “F” for left and “J” for right on the keyboard to indicate which side they believed the target object appeared on. In addition, six catch trials were interspersed to assess participants’ engagement. In these trials, a scene picture (e.g., bathroom) was presented, followed by a 2AFC question asking participants to identify the category of the scene they had just seen (e.g., kitchen or bathroom).

Data Analysis

Statistical analyses were performed using R (R Core Team, 2018) using the lme4 package (Bates et al., 2015). The analyses, scripts, and data can be accessed at <https://osf.io/m3yej>. Trials with RTs greater or less than 2 *SD* were excluded. The 2AFC performance (correct/incorrect) of each trial was modeled as a function of the scene-object congruency in that trial using logistic regression with the Bernoulli linking function.

First, we fitted three models: a full model with random slopes and intercepts for participants, a model with only random intercepts, and a model without random intercepts. For model comparison, we used Akaike information criterion (AIC), Bayesian information criterion (BIC), and chi-square tests to determine whether subsequent models should consider random effects for participants. A lower AIC or BIC value represents a better fit.

Then, we evaluated our hypotheses by comparing two models: one that included both linear and quadratic components derived from a second-order polynomial expansion of scene-object congruency (quadratic model) and another that included only a linear component (linear model). We did not constrain the direction of either the linear or quadratic components. Additionally, we evaluated a model that included only the quadratic component and compared it with the model containing both linear and quadratic components (see [Supplemental Material 5](#) for detailed results). We also analyzed and ruled out the need for a cubic component (see [Supplemental Material 4](#)). Similarly, we used AIC, BIC, and chi-square tests for model comparison.

Considering concerns about the validity of quadratic regression for identifying U-shaped relationships, we used a two-lines test (Simonsohn, 2018) to further confirm the U-shaped relationships identified in our study. Details regarding the methods and results can be found in [Supplemental Material 5](#).

Results

For 2AFC performance, we used a model with a random intercept for participants, as it provided a significantly better fit to the data compared to a model without random intercepts, $\Delta AIC = 65.57$, $\Delta BIC = 59.00$, $\chi^2(1) = 67.57$, $p < .001$, and showed no significant difference when compared to a model that included both random intercepts and slopes, $\Delta AIC = 3.62$, $\Delta BIC = 23.31$, $\chi^2(3) = 2.38$, $p = .50$.

Results indicate that in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.31$, 95% CI [0.24, 0.37], $p < .001$, $OR = 1.36$, 95% CI [1.27, 1.45]), whereas the linear component was not a significant predictor ($\beta = -0.009$, 95% CI [−0.08, 0.06], $p = .79$, $OR = 0.99$, 95% CI [0.93, 1.06]; we further discuss the effect of the linear component in [Supplemental Material 5](#)). The quadratic model provided a better fit to the data than the linear model, $\Delta AIC = 88.86$, $\Delta BIC = 82.30$, $\Delta R^2 = .03$, $\chi^2(1) = 90.86$, $p < .001$. Even after accounting for potential confounding variables (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), the quadratic component remained significant ($\beta = 0.24$, 95% CI [0.17, 0.31], $p < .001$, $OR = 1.28$, 95% CI [1.19, 1.37]), while the linear component remained nonsignificant ($\beta = -0.03$, 95% CI [−0.04, 0.11], $p = .39$, $OR = 1.03$, 95% CI [0.96, 1.12]). The quadratic model with confounding variables still provided a better fit to the data compared to the linear model with

confounding variables, $\Delta\text{AIC} = 47.19$, $\Delta\text{BIC} = 40.62$, $\Delta R^2 = .02$, $\chi^2(1) = 49.19$, $p < .001$. This suggests a robust U-shaped relationship between perceptual performance (2AFC) and scene-object congruency (see Figure 2E).

Discussion

In summary, Experiment 1a revealed a robust positive U-shaped relationship between scene-object congruency and perceptual performance. The congruency computed in Experiment 1a was based on what the actual scenes and objects were. However, participants may have inaccurately identified the scenes and objects, such as confusing an oven in the kitchen with a washing machine, which has a lower level of congruency with the kitchen. This could have led to less precise calculations of scene-object congruency. To account for this individual difference, we conducted Experiment 1b.

Experiment 1b: Exemplar Identification With Individual Differences

The purpose of Experiment 1b was to account for individual differences in scene and object comprehension and to replicate the findings of Experiment 1a. The main difference between Experiment 1b and Experiment 1a is the additional step after participants completed the identification task, where they had to fill out a questionnaire naming all scenes and objects from the stimuli. We computed a new scene-object congruency variable based on their naming responses, which was then used to predict 2AFC performance.

Method

This experiment was preregistered on the OSF at <https://osf.io/q3cn5>.

Participants

Based on the results of Experiment 1a, a minimum of 46 participants would be required to achieve 80% power. To ensure data quality, we preregistered a total of 60 participants and excluded eight participants who did not correctly complete the catch trials (see Procedure in Experiment 1a for details) from the analysis. The final data set included data from 52 participants (26 participants reported their gender as female, 26 as male; age: $M = 22.64$ years, $SD = 2.33$). All participants provided signed informed consent, had normal or corrected-to-normal vision, and were paid 15 Chinese yuan each. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

Stimuli

The stimuli used in Experiment 1b were the same as those in Experiment 1a.

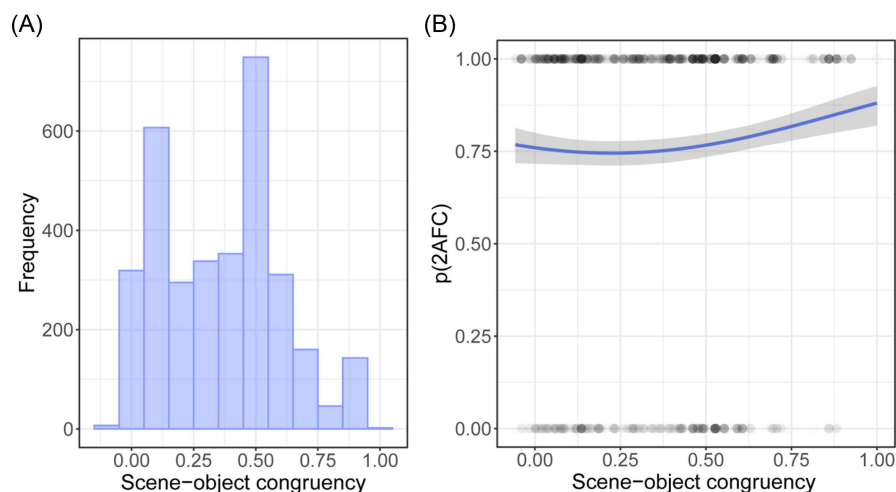
Procedure

In the first session, participants completed the same exemplar identification task as Experiment 1a. In the second session, participants were required to complete a questionnaire in which they were asked to name each pair of scene-object stimuli.

Data Analysis

Based on participants' questionnaire responses, we recalculated the scene-object congruency for each trial using ConceptNet Numberbatch (Günther et al., 2019), referred to as participant-naming-derived congruency (see Figure 3A for frequency distribution). Trials where

Figure 3
Results for Experiment 1b



Note. The frequency distribution of scene-object congruency, calculated from participants' naming responses, is depicted in (A). (B) shows the relationship between scene-object congruency and 2AFC when the congruency was calculated from participants' naming. The blue line represents the predicted second-order polynomial relationship, and the shaded area around it indicates the 95% credible intervals of the prediction. Scatter points represent the density of individual trials, where 2AFC responses were classified as correct (1) or incorrect (0). 2AFC = two-alternative forced choice. See the online article for the color version of this figure.

the named words were not included in the corpus were excluded from the analysis. Out of 3,952 trials, a total of 622 trials were excluded for this reason. We used both label-derived congruency from Experiment 1a and participant-naming-derived congruency to predict the 2AFC outcomes. The remaining data analysis procedures were consistent with Experiment 1a.

Results

We used two different measures of scene-object congruency to predict 2AFC: one based on the congruency derived from the labels and the other based on the congruency derived from the participants' naming.

For the label-derived congruency and 2AFC, the quadratic components of scene-object congruency also significantly predicted 2AFC performance ($\beta = 0.23$, 95% CI [0.15, 0.30], $p < .001$, $OR = 1.25$, 95% CI [1.16, 1.35]), whereas the linear component did not ($\beta = 0.02$, 95% CI [-0.05, 0.11], $p = .50$, $OR = 1.03$, 95% CI [0.95, 1.11]). The quadratic model fits significantly better than the linear model, $\Delta AIC = 34.12$, $\Delta BIC = 27.84$, $\Delta R^2 = 0.02$, $\chi^2(1) = 36.12$, $p < .001$. When controlling for confounding variables (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), the quadratic component remained significant ($\beta = 0.17$, 95% CI [0.09, 0.25], $p < .001$, $OR = 1.18$, 95% CI [1.09, 1.28]), while the linear component remained nonsignificant ($\beta = 0.06$, 95% CI [-0.03, 0.15], $p = .18$, $OR = 1.06$, 95% CI [0.97, 1.17]). The quadratic model with confounders fits better than the linear model, $\Delta AIC = 15.09$, $\Delta BIC = 8.81$, $\Delta R^2 = .01$, $\chi^2(1) = 17.10$, $p < .001$, supporting a U-shaped relationship.

For the participant-naming-derived congruency and 2AFC, in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.21$, 95% CI [0.13, 0.29], $p < .001$, $OR = 1.23$, 95% CI [1.13, 1.34]), while the linear component remained nonsignificant ($\beta = 0.05$, 95% CI [-0.04, 0.13], $p = .29$, $OR = 1.05$, 95% CI [0.96, 1.15]). The quadratic model fits the data significantly better than the linear model, $\Delta AIC = 24.15$, $\Delta BIC = 18.04$, $\Delta R^2 = 0.02$, $\chi^2(1) = 26.15$, $p < .001$. After controlling for confounders (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), the quadratic component remained significant ($\beta = 0.15$, 95% CI [0.06, 0.24], $p < .001$, $OR = 1.16$, 95% CI [1.06, 1.27]), while the linear component was still nonsignificant ($\beta = 0.08$, 95% CI [-0.01, 0.18], $p = .08$, $OR = 1.09$, 95% CI [0.99, 1.20]). The quadratic model with confounders again outperformed the linear model, $\Delta AIC = 9.33$, $\Delta BIC = 3.22$, $\Delta R^2 = .008$, $\chi^2(1) = 11.33$, $p < .001$, suggesting a consistent U-shaped relationship between scene-object congruency and 2AFC performance (see [Figure 3B](#)).

Discussion

Experiment 1b replicated the results of Experiment 1a, showing a robust U-shaped relationship between scene-object congruency and 2AFC, confirming the coexistence of congruency and incongruency benefits. To allow for a direct comparison with Experiment 4b, which involves a cue-present exemplar identification task using grayscale alternatives to mitigate ceiling effects, we repeated the current paradigm with a grayscale stimulus set to ensure consistency with the design of Experiment 4b.

Experiment 1c: Exemplar Identification Robust to Task Difficulty

Experiment 1c aimed to examine whether the robust U-shaped relationship between scene-object congruency and 2AFC performance could withstand increased task difficulty. To achieve this, we replaced the alternatives from Experiments 1a and 1b with grayscale versions, ensuring a wider range of scene-object congruency while maintaining consistency with the stimuli of Experiment 4b.

Method

Participants

A total of 103 participants (50 participants reported their gender as female, 53 as male; age: $M = 23.30$ years, $SD = 3.43$) participated in Experiment 1c. All participants provided signed informed consent, had normal or corrected-to-normal vision, and were paid 5 Chinese yuan each. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University. We recruited a larger sample for Experiment 1c compared to Experiments 1a and 1b due to the smaller number of stimuli. The sample size for Experiment 1c was determined based on convenience. The observed power computed by the package *SIMR* (Version 1.0.7; [Green & MacLeod, 2016](#)) was 73.50% (95% CI [66.81%, 79.48%]).

Stimuli

The primary aim of Experiment 1c was to replace the alternatives from Experiments 1a and 1b with grayscale versions while maintaining a broad range of scene-object congruency. However, for certain stimuli (e.g., bath balls), converting to grayscale reduced the discriminability between alternatives. To address this, we conducted a control experiment in which the target was extracted from the scene and placed on a white background using Adobe Photoshop, effectively removing scene information. The procedure followed that of Experiment 1a, except that grayscale alternatives were presented. Stimuli with accuracy below 1 *SD* in this control experiment might indicate poor alternative discriminability or excessively high trial difficulty. A total of 48 stimuli were selected for the Experiment 1c stimulus set. The semantic similarity scores for the remaining stimuli ranged from -0.04 to 0.89, ensuring the inclusion of both highly incongruent and highly congruent cases (see [Figure 4A](#)).

Procedure

The experimental procedure in Experiment 1c was the same as that in Experiment 1a.

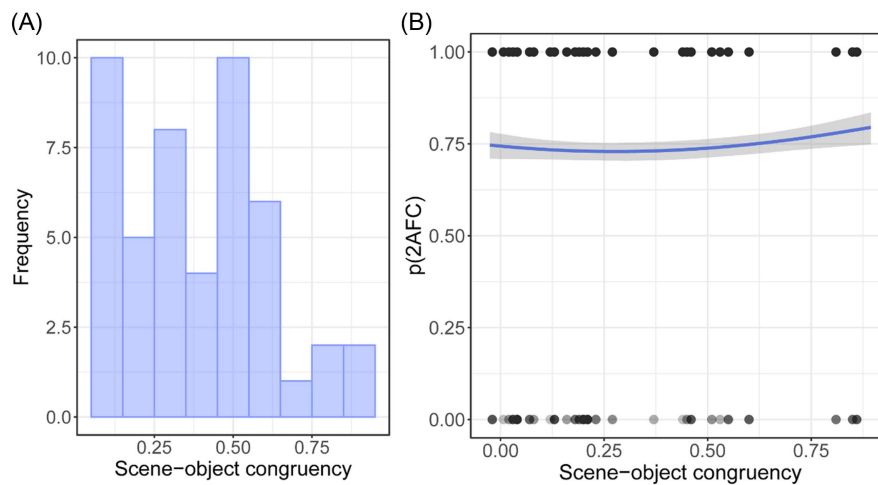
Data Analysis

The data analysis procedures were consistent with Experiment 1a.

Results

Results indicate that in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.07$, 95% CI [0.01, 0.13], $p = .03$, $OR = 1.07$, 95% CI [1.01, 1.14]), whereas the linear component was not a significant

Figure 4
Results for Experiment 1c



Note. The frequency distribution of scene-object congruency is depicted in (A). (B) shows the relationship between scene-object congruency and 2AFC. The blue line represents the predicted second-order polynomial relationship, and the shaded area around it indicates the 95% credible intervals of the prediction. Scatter points represent the density of individual trials, where 2AFC responses were classified as correct (1) or incorrect (0). 2AFC = two-alternative forced choice. See the online article for the color version of this figure.

predictor ($\beta = 0.03$, 95% CI $[-0.04, 0.10]$, $p = .42$, $OR = 1.03$, 95% CI $[0.96, 1.10]$). The quadratic model provided a better fit to the data than the linear model, $\Delta AIC = 2.85$, $\Delta BIC = -3.65$, $\Delta R^2 = .002$, $\chi^2(1) = 4.85$, $p = .03$. After controlling for potential confounders (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), the quadratic component remained significant ($\beta = 0.08$, 95% CI $[0.01, 0.14]$, $p = .024$, $OR = 1.08$, 95% CI $[1.01, 1.16]$), while the linear component remained nonsignificant ($\beta = 0.07$, 95% CI $[-0.01, 0.15]$, $p = .08$, $OR = 1.07$, 95% CI $[0.99, 1.17]$). The quadratic model with confounders still provided a better fit compared to the linear model with confounders, $\Delta AIC = 3.17$, $\Delta BIC = -3.34$, $\Delta R^2 = .002$, $\chi^2(1) = 5.17$, $p = .02$, suggesting a U-shaped relationship (see [Figure 4B](#)).

Discussion

Experiment 1c results replicated the findings of Experiments 1a and 1b, showing a robust U-shaped relationship between scene-object congruency and 2AFC performance. This U-shaped relationship remained strong even after increasing task difficulty by converting the alternatives to grayscale.

However, according to the dual-process theory ([Press et al., 2020](#)), these benefits may manifest sequentially over time. Additionally, [Krugliak et al. \(2024\)](#) found that semantic integration occurs more rapidly for objects in congruent scenes compared to incongruent scenes. These findings suggest that the observed U-shaped pattern may be influenced by variations in scene presentation times. To investigate this possibility, we conducted Experiment 2.

Experiment 2: Exemplar Identification Robust to Scene Presentation Time

Experiment 2 aims to investigate whether the previously reported U-shaped relationship is robust to differences in scene presentation

time. In Experiment 1, scenes were presented for 2,500 ms. In Experiment 2, scene presentation times are manipulated to be 1,000 ms and 4,000 ms to explore potential effects on the observed U-shaped pattern.

Method

Participants

Based on the results of Experiment 1a, a sample size of at least 46 participants would be required to achieve 80% power to test the U-shaped relationship. Our valid sample comprised 127 participants (1,000 ms group: $N = 67$, 29 participants reported their gender as female, 38 as male; age: $M = 22.48$ years, $SD = 2.39$; 4,000 ms group: $N = 60$, 26 females; age: $M = 21.52$ years, $SD = 1.77$) passing the catch trials (see Procedure in Experiment 1a for details). All participants provided signed informed consent, had normal or corrected-to-normal vision, and were paid 5 Chinese yuan each. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

Stimuli and Procedure

The stimuli and procedure in Experiment 2 were identical to Experiment 1a, with the exception of the different scene presentation times. Scene presentation times were manipulated as a between-subjects variable. Participants were assigned to two groups with scene presentation times of 1,000 ms and 4,000 ms, respectively. All other aspects of the procedure and the stimuli used remained the same as in Experiment 1a.

Data Analysis

First, we separately modeled the scene-object congruency and 2AFC relationship for the 1,000 ms and 4,000 ms groups, using the

same analytical methods as in Experiment 1a. Subsequently, we performed a combined analysis of the Experiment 1a data to explore how scene presentation time conditions moderate the association between scene-object congruency and 2AFC performance. In particular, our model incorporated interactions between scene presentation time conditions and both the linear and quadratic components.

Results

For the 2AFC performance with 1,000 ms presentation time, in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.21$, 95% CI [0.14, 0.28], $p < .001$, $OR = 1.24$, 95% CI [1.16, 1.32]), while the linear component was marginally significant ($\beta = 0.069$, 95% CI [-0.001, 0.14], $p = .05$, $OR = 1.07$, 95% CI [1.00, 1.15]). The quadratic model provided a better fit compared to the linear model, $\Delta AIC = 38.82$, $\Delta BIC = 32.29$, $\Delta R^2 = .02$, $\chi^2(1) = 40.82$, $p < .001$. After controlling for potential confounders (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), both the linear ($\beta = 0.10$, 95% CI [0.02, 0.18], $p = .01$, $OR = 1.11$, 95% CI [1.02, 1.20]) and quadratic components ($\beta = 0.15$, 95% CI [0.07, 0.22], $p < .001$, $OR = 1.16$, 95% CI [1.08, 1.24]) significantly predicted 2AFC performance. The quadratic model with confounders still outperformed the linear model with confounders, $\Delta AIC = 14.60$, $\Delta BIC = 8.06$, $\Delta R^2 = .006$, $\chi^2(1) = 16.60$, $p < .001$, demonstrating a U-shaped relationship (see [Figure 5A](#)).

Similarly, for the 2AFC performance with 4,000 ms presentation time, in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.34$, 95% CI [0.26, 0.42], $p < .001$, $OR = 1.40$, 95% CI [1.30, 1.51]), whereas the linear component was nonsignificant ($\beta = -0.01$, 95% CI [-0.09, 0.07], $p = .75$, $OR = 0.99$, 95% CI [0.91, 1.07]). The quadratic model also provided a better fit to the data than the linear

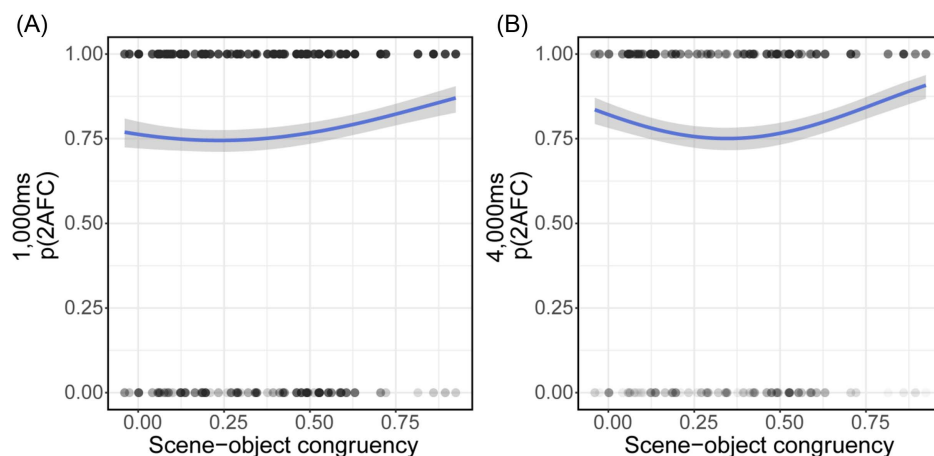
model, $\Delta AIC = 80.15$, $\Delta BIC = 73.73$, $\Delta R^2 = .04$, $\chi^2(1) = 82.15$, $p < .001$. After accounting for confounders, the quadratic component remained significant ($\beta = 0.28$, 95% CI [0.20, 0.36], $p < .001$, $OR = 1.33$, 95% CI [1.22, 1.44]), while the linear component remained nonsignificant ($\beta = 0.01$, 95% CI [-0.08, 0.10], $p = .79$, $OR = 1.01$, 95% CI [0.93, 1.11]). The quadratic model with confounders still outperformed the linear model with confounders, $\Delta AIC = 47.53$, $\Delta BIC = 41.10$, $\Delta R^2 = .02$, $\chi^2(1) = 49.53$, $p < .001$; see [Figure 5B](#).

For the combined data, the linear Congruency $\times$ Time interaction was not significant for both the 1,000 ms versus 2,500 ms comparison ($\beta = -0.04$, 95% CI [-0.15, 0.07], $p = .46$, $OR = 0.96$, 95% CI [0.86, 1.07]) and the 1,000 ms versus 4,000 ms comparison ($\beta = -0.07$, 95% CI [-0.18, 0.03], $p = .18$, $OR = 0.93$, 95% CI [0.84, 1.03]). However, the quadratic Congruency $\times$ Time interaction was significant for the 1,000 ms versus 4,000 ms comparison ($\beta = 0.12$, 95% CI [0.01, 0.22], $p = .03$, $OR = 1.12$, 95% CI [1.01, 1.25]), but not for the 1,000 ms versus 2,500 ms comparison ($\beta = 0.01$, 95% CI [-0.09, 0.11], $p = .84$, $OR = 1.01$, 95% CI [0.91, 1.12]). Model comparisons indicated that including the quadratic Congruency $\times$ Time interaction did not significantly improve model fit, $\Delta AIC = -1.14$, $\Delta BIC = -31.20$, $\Delta R^2 = .002$, $\chi^2(4) = 6.86$, $p = .14$. This suggests that the relationship between scene-object congruency and 2AFC performance did not significantly differ between the 1,000 ms and 2,500 ms conditions; however, at 4,000 ms, the quadratic effect became more pronounced, indicating a clearer U-shaped relationship.

Discussion

The results of Experiment 2 confirmed the existence of a strong positive U-shaped relationship between scene-object congruency and 2AFC performance. Importantly, this relationship was not modulated by scene presentation time, indicating its robustness across temporal conditions.

Figure 5
Results for Experiment 2



Note. (A) illustrates the relationship between scene-object congruency and 2AFC accuracy when the scene presentation time is 1,000 ms. (B) depicts the relationship between scene-object congruency and 2AFC when the scene presentation time is extended to 4,000 ms. The blue line represents the predicted second-order polynomial relationship, and the shaded area around it indicates the 95% credible intervals of the prediction. Scatter points on both figures represent the density of individual trials, where 2AFC responses were classified as correct (1) or incorrect (0). 2AFC = two-alternative forced choice. See the online article for the color version of this figure.

Moving forward, our subsequent experiments aimed to delve into the mechanisms underlying this observed U-shaped relationship. We hypothesize that this relationship may reflect the influence of scene-object congruency on both the amount of processing (attention) and processing efficiency. In the following two experiments, we systematically investigate these two aspects.

Experiment 3: Change Detection Task

Experiment 3 aims to investigate whether scene-object congruency influences object perception by affecting attention and, consequently, processing amount. To test this, Experiment 3 employed a change detection paradigm, focusing primarily on attentional processes. We used a vector-space model of semantics to quantify congruency. The stimulus set used was the same as in Experiment 1, which included a wide range of congruencies. We hypothesized an inverted U-shaped relationship between scene-object congruency and change detection RT.

Method

This experiment was preregistered on the OSF at <https://osf.io/ng9ec>.

Participants

Twenty-five participants (17 participants reported their gender as female, eight as male; $M = 21.84$ years, $SD = 1.69$ years) participated in Experiment 3, each receiving 5 Chinese yuan for their participation. All participants were undergraduate and graduate students of Sun Yat-sen University who provided signed informed consent and had normal or corrected-to-normal vision. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

Sample size was determined to reach at least 80% power based on the quadratic effect size in the pilot experiment with the package *SIMR* (Version 1.0.7; Green & MacLeod, 2016). The suggested sample size was 25 individuals.

Stimuli

We used the same stimulus set as in Experiment 1, comprising 90 real-world scene images. The congruency values ranged from -0.04 to 0.92 . Each stimulus has two versions, one with and one without a target object.

Apparatus

The experiment was programmed and carried out using PsychoPy (Version v2021.2.3; Peirce et al., 2019). Stimuli were presented on a 23-inch LCD monitor (1920×1080 resolution, 60 Hz refresh rate) at a viewing distance of approximately 57 cm.

Procedure

This experiment was adapted from the classic change detection “flicker” task (Rensink et al., 1997), as shown in Figure 6A. Participants were required to detect changes between successive displays of a scene. Each trial began with a 500 ms blank screen,

followed by a small button labeled “Go” appearing in the center of the screen. Participants needed to click the button with the mouse to start the scene presentation, ensuring that they were always fixating on the center when the stimuli appeared. The stimuli presentation started with a 250-ms scene with the object present, followed by a 100 ms blank screen, and then a 250-ms scene with the object absent. This sequence was presented repeatedly until the participant presses the space bar to indicate that they have detected the change or until the trial has been repeated 13 times. After the participant responded, a gray square of the same size as the scene image was presented, and the mouse appeared from a hidden state. Participants needed to click on the location corresponding to the change in the object. Only responses with clicking positions correctly falling on the shape of the object were considered valid response data, ensuring that the participant had truly detected the change rather than responding randomly.

Data Analysis

For each stimulus image, we generated binary segmentation masks to delineate the target objects (where the target objects were represented in black and the remaining areas were represented in white), which allowed us to calculate the boundaries of the target object. Only trials in which the localization fell within the boundaries of the target object were considered successful in detecting change.

Data analysis closely mirrored Experiment 1, with the key difference being the dependent variable—change detection RT, which was modeled using linear regression. We focused our RT analysis only on trials in which change detection was successful. We $\log_{10}$ -transformed RT and localization error to improve normality.

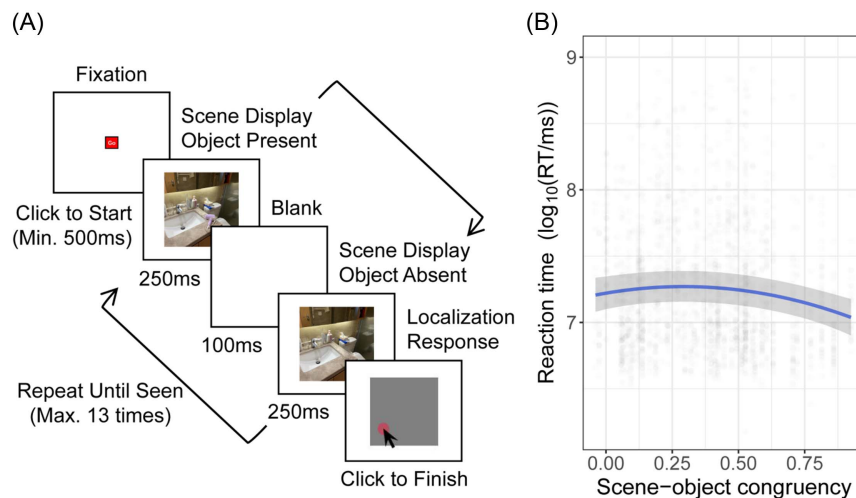
Results

Overall, change detection RTs were well within the maximum of 7.8 s ($M = 1,487$ ms, $SD = 892$). 11.69% of the total trials were excluded from the RT analyses due to failure of change detection.

For the analysis of RTs, a model with a random intercept for participants was used, as it provided a significantly better fit to the data compared to both a model without random intercepts, $\Delta AIC = 484.69$, $\Delta BIC = 479.09$, $\chi^2(1) = 486.69$, $p < .001$, and a model that included both random intercepts and slopes, $\Delta AIC = 5.87$, $\Delta BIC = 22.69$, $\chi^2(3) = 0.13$, $p = .99$. This suggests that while the shape of the relationship between RT and scene-object congruency remained consistent across participants, the baseline level of RT (intercept) varied across participants.

In the quadratic model, both the linear ($\beta = -0.01$, 95% CI $[-0.02, -0.004]$, $p = .002$) and quadratic components of scene-object congruency ($\beta = -0.02$, 95% CI $[-0.03, -0.01]$, $p < .001$) significantly predicted RT. The quadratic model provided a better fit to the data than the linear model, $\Delta AIC = 26.41$, $\Delta BIC = 20.80$, $\Delta R^2 = .01$, $\chi^2(1) = 28.41$, $p < .001$. After controlling for potential confounding variables (saliency, size, center proximity, complexity, and familiarity; see Supplemental Material 3 for detailed results), both the linear ($\beta = -0.01$, 95% CI $[-0.02, -0.001]$, $p = .03$) and quadratic components ($\beta = -0.01$, 95% CI $[-0.02, -0.007]$, $p < .001$) remained significant. The quadratic model with confounding variables still showed a better fit than the linear model with confounders, $\Delta AIC = 13.10$, $\Delta BIC = 7.49$, $\Delta R^2 = .005$, $\chi^2(1) = 15.10$, $p < .001$, showing an inverted U-shaped pattern of scene-object congruency and RTs (see Figure 6B).

Figure 6
Design and Results for Experiment 3



Note. The experimental procedure of Experiment 3 is shown in (A). Participants were asked to detect changes between successive presentations of a scene. The participant had to click the “Go” button to initiate the scene presentation, which consisted of a 250-ms scene with the object present, followed by a 100 ms blank screen, and then a 250-ms scene without the object. This sequence was repeated until the participant pressed the space bar to indicate change detection or until the trial was repeated 13 times. After responding, participants clicked on the location on a gray square corresponding to the changed object. (B) presents the RT results for Experiment 3, with scene-object congruency values on the *x*-axis and RT on the *y*-axis. The blue line represents the predicted second-order polynomial relationship, and the shaded area around the blue line indicates the 95% credible intervals of the prediction. Scatter points represent the RT for each trial of each participant. RT = reaction time; Min = minimum; Max = maximum. See the online article for the color version of this figure.

Discussion

Experiment 3 provides robust evidence for an inverted U-shaped relationship between scene-object congruency and RT in the change detection task. Our results extend previous work by examining a broader range of congruencies and confirming both incongruency (Hollingworth & Henderson, 2000; LaPointe et al., 2013; Mack et al., 2017; Spaak et al., 2022) and congruency benefits (Rensink et al., 1997) in the change detection task. This finding suggests that both congruent and incongruent contexts enhance object perception by capturing attention. Experiment 4, in contrast, uses a paradigm that emphasizes processing efficiency to further investigate the effect of scene-object congruency on object perception.

Experiment 4a: Cue-Present Exemplar Identification Task

To investigate whether the previously observed U-shaped relationship reflects the advantages of processing efficiency, we conducted Experiment 4a. Similar to previous experiments, participants in Experiment 4a had to identify the target object from two exemplars. However, in contrast to previous experiments, a valid attentional cue was presented before the target object, directing participants' gaze to the target object. In other words, the attentional advantage was no longer in effect. We hypothesized that by disrupting one end of the U-shaped relationship, there would be a positive linear relationship between congruency and identification performance. Since the attentional cue significantly reduces the difficulty of the task, we used

the same stimulus set in Experiment 1c whose alternatives could be presented in grayscale to increase the difficulty of the task.

Method

Participants

Ninety-four participants (52 participants reported their gender as female, 42 as male; $M = 23.03$ years, $SD = 3.13$ years) participated in Experiment 4a, each receiving 5 Chinese yuan for their participation. All participants provided signed informed consent and had normal or corrected-to-normal vision. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

The sample size of Experiment 4a was determined for convenience. The observed power computed by the package *SIMR* (Version 1.0.7; Green & MacLeod, 2016) was 99.00% (95% CI [98.17%, 99.52%]).

Stimuli

The stimuli in Experiment 4a were identical to Experiment 1c.

Procedure

We adopted a paradigm similar to that of Chen et al. (2022). As shown in Figure 7A, each trial began with a fixation cross (1,300–1,500 ms, randomly selected from a uniform distribution). A scene image (without the target object) was then displayed for 500 ms.

Then, a red dot cue indicated the location (500 ms) where the target would appear. A target object was then presented at the red dot location for 83 ms. The scene image remained on the screen for another 500 ms after the object disappeared. After a 200-ms blank interval, participants made an exemplar identification task: Two object exemplars were presented on the screen—one from the previously viewed scene and the other representing a different exemplar but of the same category. The object choices were intentionally presented in grayscale to increase task difficulty. Participants were required to perform a 2AFC by pressing the “F” or “J” key to select the alternative on the left or right side of the screen, respectively.

Data Analysis

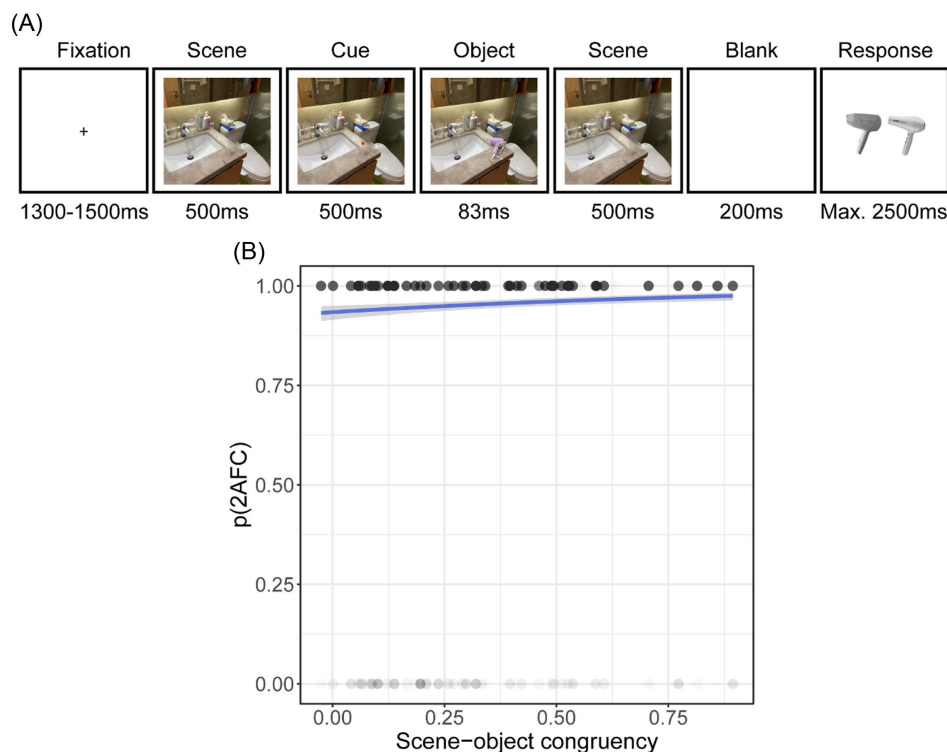
The data analysis procedures were consistent with Experiment 1a.

Results

Results indicate that in the linear model, the linear components of scene-object congruency significantly predicted 2AFC

performance ($\beta = 0.29$, 95% CI [0.16, 0.42], $p < .001$, $OR = 1.34$, 95% CI [1.17, 1.53]). However, in the quadratic model, the quadratic component was not a significant predictor ($\beta = -0.02$, 95% CI [-0.15, 0.10], $p = .70$, $OR = 0.98$, 95% CI [0.86, 1.10]). Model comparisons showed that adding the quadratic component did not significantly improve model fit, $\Delta AIC = -1.87$, $\Delta BIC = -8.26$, $\Delta R^2 = -.0005$, $\chi^2(1) = 0.13$, $p = .72$. Even after accounting for potential confounding variables (saliency, size, center proximity, complexity, and familiarity; see [Supplemental Material 3](#) for detailed results), the linear component remained significant in the linear model ($\beta = 0.16$, 95% CI [0.02, 0.31], $p = .03$, $OR = 1.18$, 95% CI [1.02, 1.36]), while the quadratic component in the quadratic model remained nonsignificant ($\beta = 0.07$, 95% CI [-0.07, 0.21], $p = .33$, $OR = 1.07$, 95% CI [0.93, 1.23]). The quadratic model with confounding variables also did not significantly improve the model fit, $\Delta AIC = -1.08$, $\Delta BIC = -7.47$, $\Delta R^2 = .0006$, $\chi^2(1) = 0.92$, $p = .34$. These findings suggest a linear relationship between scene-object congruency and 2AFC performance in the presence of attentional cues, without evidence of a U-shaped effect (see [Figure 7B](#)).

Figure 7
Design and Results for Experiment 4a



Note. The experimental procedure of Experiment 4a is shown in (A). Each trial started with a fixation cross. A scene image (without the target object) appeared for 500 ms, followed by a red dot cue indicating the target location for 500 ms. The target object was presented at the red dot location for 83 ms. After the object disappeared, the scene image remained for a further 500 ms, followed by a 200-ms blank interval. Participants were asked to complete a 2AFC task to identify the previously presented exemplar. (B) shows the relationship between scene-object congruency and 2AFC. The blue line represents the predicted linear relationship, and the shaded area around it indicates the 95% credible intervals of the prediction. Scatter points represent the density of individual trials, where 2AFC responses were classified as correct (1) or incorrect (0). 2AFC = two-alternative forced choice; Max = maximum. See the online article for the color version of this figure.

Discussion

The findings from Experiment 4a reveal that when participants' attention is effectively directed to the target object by the attentional cue, the previously observed U-shaped relationship between congruency and identification performance in Experiment 1c no longer holds. Instead, a linear relationship emerges. To further validate this linear pattern between congruency and 2AFC performance under cued conditions, Experiment 4b replicates the procedure with a different stimulus set.

Experiment 4b: Cue-Present Exemplar Identification Across Stimulus Sets

Building on the findings from Experiment 4a, Experiment 4b seeks to further investigate the relationship between scene-object congruency and 2AFC performance in the presence of attentional cues. A new stimulus set has been introduced, carefully balanced across three levels of congruency: highly congruent, moderately congruent, and incongruent. This balanced design enables a more precise assessment of the congruency effect while reducing the potential impact of confounding variables.

Method

Participants

Seventy-three participants (56 participants reported their gender as female, 17 as male; age: $M = 20.68$ years, $SD = 1.82$) participated in Experiment 4b and were paid 8 Chinese yuan each. All participants provided signed informed consent and had normal or corrected-to-normal vision. All experimental procedures in our present study were approved by the institutional review board of the Department of Psychology, Sun Yat-sen University.

We used the PANGAEA software (Westfall, 2015) to calculate the sample size. Based on the pilot study, the estimated effect size (Cohen's $d = 0.37$) for a one-factor within-subject design indicates that, with 95% power and a significance level of 0.05, a minimum sample size of 73 participants was required.

Stimuli

In Experiment 4b, a new stimulus set was used because of the inherent difficulties associated with using the previous stimulus set in a cue-present identification task. Typically, grayscale alternatives are introduced in 2AFC to prevent ceiling effects, but in the previous stimulus set, some grayscale versions of the alternatives were excessively difficult to identify.

The new stimulus set is balanced across three levels: highly congruent, moderately congruent, and incongruent, thereby minimizing the influence of confounding factors. The three congruency levels were determined based on semantic similarity calculated using ConceptNet Numberbatch (Günther et al., 2019) word vectors (see Supplemental Material 2.3 for all scene-object pairings). Overall, the range of congruency values for this stimulus set, spanning from -0.02 to 0.86 , is similar to the range used in the previous stimulus set (see Figure 8A). Objects were inserted into the scene images using Adobe Photoshop,¹ maintaining consistent object sizes across the three congruency levels. The set includes 16 scene categories: orchard, soccer field, bar, gym, bathroom, basketball court, library, balcony, kitchen, living room,

coffee shop, bakery, office, study, park, restaurant, beach, and classroom. Each scene category is matched with three objects corresponding to three levels of congruency (see Figure 8B). Three exemplars were generated for each scene category and three for each object, resulting in 324 unique stimuli (see Figure 8C).

Data Analysis

We conducted two types of analyses on the data. First, based on the way the stimulus set was balanced, stimuli were categorized into three levels: highly congruent, moderately congruent, and incongruent. To examine the differences in 2AFC accuracy across these levels, we performed both frequentist and Bayesian repeated measures analysis of variance using JASP (JASP Team, 2020), followed by post hoc pairwise comparisons with Bonferroni correction. If the assumption of sphericity was violated (Mauchly's test, $p < .05$), the Greenhouse–Geisser correction was applied to adjust all degrees of freedom.

Second, similar to previous experiments, we treated congruency as a continuous variable and conducted regression analyses as Experiment 1a.

Results

When congruency was categorized into three levels, as shown in Figure 8D, for 2AFC accuracy, we observed a significant main effect of congruency level, $F(1.82, 130.94) = 5.36$, $p = .007$, $\eta_p^2 = 0.07$, $BF_{10} = 5.22$. Post hoc pairwise comparisons revealed that accuracy was significantly higher in the highly congruent condition than in the moderately congruent, $t(72) = 3.11$, $p = .007$, Cohen's $d = 0.29$, $BF_{10} = 7.88$, and incongruent conditions, $t(72) = 2.43$, $p = .049$, Cohen's $d = 0.22$, $BF_{10} = 6.16$, whereas there was no significant difference between the moderately congruent and incongruent conditions, $t(72) = -0.68$, $p = 1.00$, Cohen's $d = -0.06$, $BF_{10} = 0.15$.

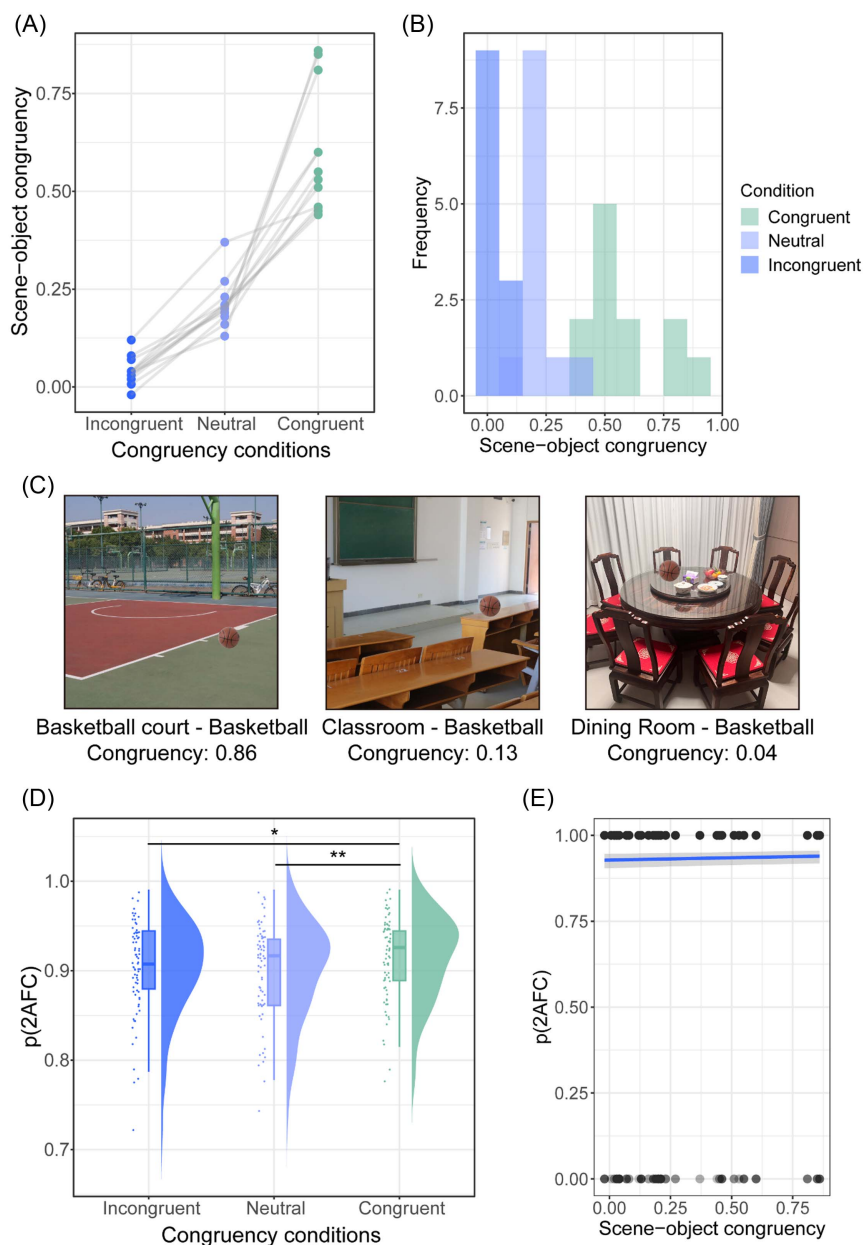
We also treated congruency as a continuous variable. Results indicate that in the linear model, the linear component of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.06$, 95% CI [0.004, 0.11], $p = .03$, $OR = 1.06$, 95% CI [1.00, 1.11]). In the quadratic model, the quadratic component was not a significant predictor ($\beta = 0.01$, 95% CI [-0.03, 0.12], $p = .65$, $OR = 1.01$, 95% CI [0.96, 1.07]). Model comparisons indicated that including the quadratic component did not significantly improve the model fit, $\Delta AIC = -1.80$, $\Delta BIC = -9.87$, $\Delta R^2 = .001$, $\chi^2(1) = 0.20$, $p = .65$. This suggests a linear relationship between perceptual performance (2AFC) and scene-object congruency with the presence of an attentional cue (see Figure 8E).

Discussion

The results of Experiment 4b replicate the findings from Experiment 4a: When attentional cues are present, the relationship between congruency and 2AFC performance is linear. In other words, the advantage of incongruency disappears. These findings suggest that the incongruency benefit observed at the other end of the U-shape in previous experiments is primarily driven by congruency-based attention, which in turn influences the amount of

¹ We did not use this stimulus set to replicate Experiments 1 and 2 because edited images are not suitable for cue-absent identification tasks. The editing traces may attract attention and cause confounding.

Figure 8
Design and Results for Experiment 4b



Note. (A) illustrates the scene-object congruency for the stimuli used in Experiment 4b. The x-axis represents three congruency levels: congruent, neutral (moderately congruent), and incongruent. Each line signifies the congruency values corresponding to a specific object across the three congruency levels. (B) represents the frequency distribution of scene-object congruency. The green, light purple, and deep purple distributions represent stimuli originally categorized as congruent, neutral, and incongruent, respectively. (C) shows examples of the scene-object pairings used in Experiment 4b, along with their congruency scores. (D) and (E) show the results of Experiment 4. In (D), the x-axis denotes the three congruency levels, while the y-axis represents 2AFC accuracy. Each data point represents the average accuracy for each participant at the respective congruency level. (E) shows the relationship between scene-object congruency and 2AFC. The blue line represents the predicted linear relationship, and the shaded area around it indicates the 95% credible intervals of the prediction. Scatter points represent the density of individual trials, where 2AFC responses were classified as correct (1) or incorrect (0). 2AFC = two-alternative forced choice. See the online article for the color version of this figure.

* $p < .05$. ** $p < .01$.

processing and leads to perceptual benefits. In the context of such brief stimulus processing times, identification performance on highly congruent trials is significantly higher than that on moderately congruent and incongruent conditions, demonstrating the processing efficiency advantage of congruent objects, consistent with previous findings (e.g., Chen et al., 2022).

Reanalysis of Spaak et al. (2022)

Given that Experiments 1 to 4a used the same set of stimuli, the possibility remains that the observed U-shape was a quirk of specific stimuli. To address this, we reanalyzed the data from Spaak et al. (2022), which had a similar experimental design to ours, and compared perceptual performance between congruent and incongruent conditions. We used ConceptNet Numberbatch (Günther et al., 2019) to compute scene-object congruency for the stimuli they used and then repredicted 2AFC.

Method

The data are publicly available via the Donders Repository at <https://doi.org/10.34973/x6fy-4q26>.

In their original study, 100 participants participated in a change detection task, which was consistent with the procedure used in our Experiment 3. Another 194 participants participated in an exemplar identification task, which was consistent with the procedure used in our Experiment 1a. The stimuli employed in Spaak et al.'s (2022) original experiment were sourced from the SCEGRAM stimulus database (Öhlschläger & Vö, 2017), specifically utilizing all 62 scenes categorized under the "CON" (congruent) and "SEM" (incongruent) conditions.

Data Analysis

In the original study by Spaak et al. (2022), both RT and 2AFC performance were compared between congruent and incongruent conditions. Instead of binary classification of individual trials, we employed ConceptNet Numberbatch (Günther et al., 2019) to generate semantic vectors for scene and object words. Subsequently, we calculated the semantic similarity between these vectors. This semantic similarity served as a quantitative measure of scene-object congruency to predict RT and 2AFC, respectively (see Figure 9A for frequency distribution). Subsequent data analysis procedures remained consistent with those used in Experiments 1 and 3.

Results

For change detection RT in Spaak et al.'s (2022) Experiment 1, results showed both the linear ($\beta = 0.03$, 95% CI [0.01, 0.05], $p < .001$) and quadratic components ($\beta = -0.03$, 95% CI [-0.04, -0.02], $p < .001$) of scene-object congruency significantly predicted RT. The quadratic model provided a better fit to the data than the linear model, $\Delta AIC = 65.71$, $\Delta BIC = 59.07$, $\Delta R^2 = .01$, $\chi^2(1) = 67.71$, $p < .001$, suggesting an inverted U-shaped relationship between scene-object congruency and RT (see Figure 9B).

For 2AFC in Spaak et al.'s (2022) Experiment 2, results indicate that in the quadratic model, the quadratic components of scene-object congruency significantly predicted 2AFC performance ($\beta = 0.04$, 95% CI [0.009, 0.07], $p = .01$, $OR = 1.04$, 95% CI [1.01,

1.07]), whereas the linear component was not significant ($\beta = -0.06$, 95% CI [-0.13, 0.02], $p = .12$, $OR = 0.94$, 95% CI [0.88, 1.02]). The quadratic model provided a better fit to the data than the linear model, $\Delta AIC = 4.61$, $\Delta BIC = -2.07$, $\Delta R^2 = .003$, $\chi^2(1) = 6.61$, $p = .01$, suggesting a U-shaped relationship between scene-object congruency and perceptual performance (see Figure 9C).

Discussion

The reanalysis by Spaak et al. (2022) confirms that when scene-object congruency is treated as a continuous variable, the relationship with RT follows an inverted U-shaped pattern and the relationship with 2AFC follows a U-shaped pattern. This replication of findings from previous experiments suggests that the U-shaped results observed earlier can be generalized to different stimulus sets.

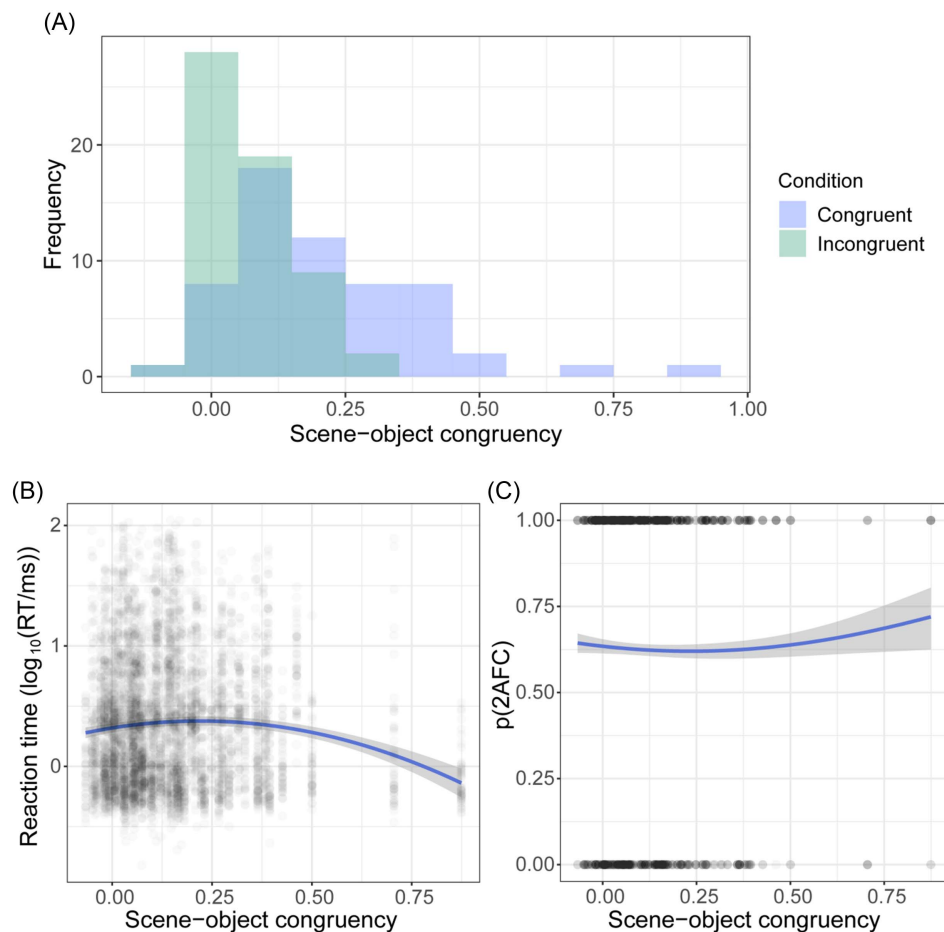
General Discussion

In summary, our study investigates the relationship between scene-object congruency and object perception using multiple paradigms and a wide range of congruency values quantified by a vector-space semantic model. Experiment 1 demonstrated a positive U-shaped relationship between congruency and performance in object identification, a relationship robust to scene presentation time (Experiment 2) and generalizable across different stimulus sets (reanalysis of the previous data set). To further explore the mechanisms underlying this U-shaped pattern, Experiment 3 used a change detection task, emphasizing the attention process, revealing an inverted U-shaped relationship between congruency and change detection RTs. Experiment 4, employing an identification task with attentional cues, aimed to isolate processing efficiency. The results revealed only congruency benefits, with the disappearance of incongruency benefits at the opposite end of the U-shaped curve. Together, our findings unveil the coexistence of congruency and incongruency benefits. These benefits may reflect a U-shaped enhancement of attention (and hence processing amount) by congruency, as well as processing efficiency advantages specific to high congruency.

Attention (and Processing Amount)

One of the mechanisms to explain the U-shaped relationship we observed is the U-shaped relationship between congruency and attention, which in turn affects the processing amount. This is evident in both the change detection task and the exemplar identification task. Specifically, we found an inverted U-shaped relationship between congruency and RT in change detection (Experiment 3 and reanalysis of Spaak et al.'s, 2022, Experiment 1). Given that attention capture processes are considered crucial for the detection and localization of changing objects (LaPointe et al., 2013), this finding likely reflects a U-shaped relationship between congruency and attention. In the exemplar identification task, when participants were free to view the scenes, congruency showed a U-shaped relationship with object perception (Experiments 1 to 2). However, when participants' attention was directed to the target location by cues, thereby eliminating the influence of congruency-based attention, the U-shaped relationship disappeared (Experiment 4). This finding indirectly suggests that the previously observed U-shaped relationship may reflect, at least in part, the relationship between congruency and attention.

Figure 9
Results for Reanalysis of *Spaak et al. (2022)*



Note. (A) depicts the frequency distribution of scene-object congruency, recalculated for the stimuli used by *Spaak et al. (2022)*. The light purple distribution represents stimuli originally categorized as congruent with the scene, while the light green distribution represents stimuli categorized as incongruent with the scene. (B) and (C) present the reanalyzed results of *Spaak et al.'s (2022)* Experiment 1 and Experiment 2, respectively. The x-axis in both figures represents the recalculated scene-object congruency. The blue line represents the predicted second-order polynomial relationship, with the shaded area around it representing the 95% credible intervals of the prediction. Scatter points represent individual trial data. 2AFC = two-alternative forced choice; RT = reaction time. See the online article for the color version of this figure.

Our findings reconcile previous contradictions and suggest that these contradictions may actually converge on a U-shaped relationship between congruency and attention. Previous research using change detection tasks has predominantly found incongruency benefit (e.g., *Hollingworth & Henderson, 2000; LaPointe et al., 2013*). However, *Rensink et al. (1997)* found that “interesting” objects, which are crucial for scene understanding, showed better performance in change detection tasks, suggesting a congruency benefit. Our study supports both perspectives by revealing that both congruency and incongruency benefits coexist in change detection tasks, indicating a U-shaped relationship between congruency and attention. Recent work by *Damiano et al. (2024)* also confirms this U-shaped pattern, where both highly congruent and highly incongruent objects attract more fixations than moderately congruent ones.

Why does the relationship between scene-object congruency and attention show a U-shaped pattern? Scene context information is known to guide attention to the most informative regions (*Kaiser et al., 2019; Peelen & Kastner, 2014; Vö et al., 2019; Wolfe & Horowitz, 2017; Wu et al., 2014*). On the one hand, objects that are incongruent with scenes are informative because they violate our expectations (*Press et al., 2020*). On the other hand, highly congruent objects may carry much information about a particular scene. While direct research on this is lacking, two analogous concepts have emerged. Some objects are considered crucial for scene categorization and have been identified as diagnostic objects, providing insight into the “what” information of the scene (*Biederman et al., 1982; Friedman, 1979; Greene, 2013; MacEvoy & Epstein, 2011*). For example, the presentation of a single diagnostic object for as little as 50 ms is sufficient for correct scene categorization (*Wiesmann & Vö, 2023*). In addition,

recent research has introduced the concept of anchor objects, which refer to objects that provide spatial information about *where* other objects might be (Boettcher et al., 2018; Draschkow & Vö, 2017; Vö, 2021; Vö et al., 2019). For example, in a bathroom scene, a toilet can predict the location of a toilet brush. Anchor objects have been found to effectively guide attention (Boettcher et al., 2018; Helbing et al., 2022). Although diagnostic objects, anchor objects, and highly scene-object congruent objects have different definitions, they are clearly highly overlapping, typically representing specific and frequently occurring objects in scenes (Wiesmann & Vö, 2023). Future research could systematically quantify the degree of overlap between these three types of objects. Overall, the potential for overlap suggests that highly congruent and incongruent objects may provide information from different perspectives, contributing to advantages in object attention and perception.

Processing Efficiency

Another possible mechanism explaining the relationship between scene-object congruency and object perception is processing efficiency. Our findings revealed a congruency benefit in identifying a briefly presented (83 ms) target object (Experiment 4), aligning with prior research (Chen et al., 2022). Such processing efficiency advantages are typically attributed to scene gist priors, which aid in the generation of predictions about potential objects (Chen et al., 2022; Krugliak et al., 2024; Press et al., 2020).

Notably, this benefit was specific to highly congruent objects, whereas moderately congruent and incongruent objects showed comparable identification performance (Experiment 4b). The lack of benefit for moderately congruent objects may stem from their weaker binding with scenes. For instance, while observers may not be surprised to see a vase in a kitchen, few would predict its presence beforehand. Our results suggest that scene-generated predictions of objects are very limited and primarily restricted to objects that are highly congruent with the scene. Since highly congruent objects are rare compared to the numerous moderately congruent objects, this reduction in redundancy highlights the efficiency of the visual system.

Opposing Process Theory

In addition, our study provides empirical evidence in support of the opposing process theory (Press et al., 2020). The opposing process theory posits that our perceptions are initially biased toward prior expectations. It is only when the unexpected events are sufficiently surprising that the processing of the unexpected events is upweighted. Our results are consistent with this framework: Both congruency and incongruency benefits coexist. Our quantification of continuous congruency is also consistent with the theory's suggestion that "surprise" needs to be quantified. However, due to the temporal insensitivity of our behavioral experiments, we were unable to provide evidence for the temporal sequence of the two stages proposed by the opponent-process theory. Future research could use time-sensitive experiments to investigate the temporal sequence of perceptual advantages at both ends of the U-shaped pattern.

Conclusion

In summary, we have comprehensively investigated the relationship between scene-object congruency and object perception.

We introduced the concept of congruency as a continuous variable, distinguishing between highly congruent, moderately congruent, and incongruent objects. Our results revealed a positive U-shaped relationship between scene-object congruency and object perception, representing the interplay between processing amount and efficiency advantages. Utilizing advances in computational modeling, our study offers a new perspective to reconcile previous contradictions and unify the benefits of scene-object congruency and incongruency in object perception.

Constraints on Generality

Our study reveals how scene-object congruency influences object perception. The stimuli in our experiments include both unedited real scene stimulus sets and stimulus sets edited with image editing software. We also reanalyzed data from stimulus sets used in previous studies that were well controlled. We expect the generalizability of our results to other similar scene-object stimulus sets. Our participants consisted of undergraduate and graduate students from Sun Yat-sen University, and we expect that these results can be replicated with participants from similar subject pools. The calculation of scene-object congruency in our experiments was based on the Chinese version of ConceptNet Numberbatch (Günther et al., 2019). We expect that our results can be extended as long as congruency is computed on the basis of the language commonly used by the participants. Furthermore, the results of this study rule out the influence of scene presentation time. We have no reason to believe that the results depend on other characteristics of the participants, materials, or context.

References

- Anzulewicz, A., Asanowicz, D., Windey, B., Paulewicz, B., Wierchoń, M., & Cleeremans, A. (2015). Does level of processing affect the transition from unconscious to conscious perception? *Consciousness and Cognition*, 36, 1–11. <https://doi.org/10.1016/j.concog.2015.05.004>
- Bar, M. (2004). Visual objects in context. *Nature Reviews Neuroscience*, 5(8), 617–629. <https://doi.org/10.1038/nrn1476>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Biederman, I., Mezzanotte, R. J., & Rabinowitz, J. C. (1982). Scene perception: Detecting and judging objects undergoing relational violations. *Cognitive Psychology*, 14(2), 143–177. [https://doi.org/10.1016/0010-0285\(82\)90007-X](https://doi.org/10.1016/0010-0285(82)90007-X)
- Boettcher, S. E., Draschkow, D., Dienhart, E., & Vö, M. L. H. (2018). Anchoring visual search in scenes: Assessing the role of anchor objects on eye movements during visual search. *Journal of Vision*, 18(13), Article 11. <https://doi.org/10.1167/18.13.11>
- Bonitz, V. S., & Gordon, R. D. (2008). Attention to smoking-related and incongruous objects during scene viewing. *Acta Psychologica*, 129(2), 255–263. <https://doi.org/10.1016/j.actpsy.2008.08.006>
- Boyce, S. J., Pollatsek, A., & Rayner, K. (1989). Effect of background information on object identification. *Journal of Experimental Psychology: Human Perception and Performance*, 15(3), 556–566. <https://doi.org/10.1037/0096-1523.15.3.556>
- Chai, X. J., Ofen, N., Jacobs, L. F., & Gabrieli, J. D. (2010). Scene complexity: Influence on perception, memory, and development in the medial temporal lobe. *Frontiers in Human Neuroscience*, 4, Article 21. <https://doi.org/10.3389/fnhum.2010.00021>
- Chen, L., Cichy, R. M., & Kaiser, D. (2022). Semantic scene-object consistency modulates N300/400 EEG components, but does not automatically

- facilitate object representations. *Cerebral Cortex*, 32(16), 3553–3567. <https://doi.org/10.1093/cercor/bhab433>
- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204. <https://doi.org/10.1017/S0140525X12000477>
- Cornelissen, T. H., & Vö, M. L. H. (2017). Stuck on semantics: Processing of irrelevant object-scene inconsistencies modulates ongoing gaze behavior. *Attention, Perception & Psychophysics*, 79(1), 154–168. <https://doi.org/10.3758/s13414-016-1203-7>
- Damiano, C., Leemans, M., & Wagemans, J. (2024). Exploring the semantic-inconsistency effect in scenes using a continuous measure of linguistic-semantic similarity. *Psychological Science*, 35(6), 623–634. <https://doi.org/10.1177/09567976241238217>
- Davenport, J. L. (2007). Consistency effects between objects in scenes. *Memory & Cognition*, 35(3), 393–401. <https://doi.org/10.3758/BF03193280>
- Davenport, J. L., & Potter, M. C. (2004). Scene consistency in object and background perception. *Psychological Science*, 15(8), 559–564. <https://doi.org/10.1111/j.0956-7976.2004.00719.x>
- de Lange, F. P., Heilbron, M., & Kok, P. (2018). How do expectations shape perception? *Trends in Cognitive Sciences*, 22(9), 764–779. <https://doi.org/10.1016/j.tics.2018.06.002>
- Draschkow, D., & Vö, M. L. H. (2017). Scene grammar shapes the way we interact with objects, strengthens memories, and speeds search. *Scientific Reports*, 7(1), Article 16471. <https://doi.org/10.1038/s41598-017-16739-x>
- Fize, D., Cauchois, M., & Fabre-Thorpe, M. (2011). Humans and monkeys share visual representations. *Proceedings of the National Academy of Sciences of the United States of America*, 108(18), 7635–7640. <https://doi.org/10.1073/pnas.1016213108>
- Friedman, A. (1979). Framing pictures: The role of knowledge in automatized encoding and memory for gist. *Journal of Experimental Psychology: General*, 108(3), 316–355. <https://doi.org/10.1037/0096-3445.108.3.316>
- Friston, K. (2005). A theory of cortical responses. *Philosophical Transactions of the Royal Society of London Series B: Biological Sciences*, 360(1456), 815–836. <https://doi.org/10.1098/rstb.2005.1622>
- Green, P., & MacLeod, C. J. (2016). SIMR: An R package for power analysis of generalized linear mixed models by simulation. *Methods in Ecology and Evolution*, 7(4), 493–498. <https://doi.org/10.1111/2041-210X.12504>
- Greene, M. R. (2013). Statistics of high-level scene context. *Frontiers in Psychology*, 4, Article 777. <https://doi.org/10.3389/fpsyg.2013.00777>
- Günther, F., Rinaldi, L., & Marelli, M. (2019). Vector-space models of semantic representation from a cognitive perspective: A discussion of common misconceptions. *Perspectives on Psychological Science*, 14(6), 1006–1033. <https://doi.org/10.1177/1745691619861372>
- Hayes, T. R., & Henderson, J. M. (2021). Looking for semantic similarity: What a vector-space model of semantics can tell us about attention in real-world scenes. *Psychological Science*, 32(8), 1262–1270. <https://doi.org/10.1177/0956797621994768>
- Helbing, J., Draschkow, D., & Vö, M. L.-H. (2022). Auxiliary scene-context information provided by anchor objects guides attention and locomotion in natural search behavior. *Psychological Science*, 33(9), 1463–1476. <https://doi.org/10.1177/09567976221091838>
- Hollingworth, A., & Henderson, J. M. (2000). Semantic informativeness mediates the detection of changes in natural scenes. *Visual Cognition*, 7(1–3), 213–235. <https://doi.org/10.1080/135062800394775>
- JASP Team. (2020). JASP (Version 0.12.0) [Computer software]. <https://jasp-stats.org/>
- Kaiser, D., Quek, G. L., Cichy, R. M., & Peelen, M. V. (2019). Object vision in a structured world. *Trends in Cognitive Sciences*, 23(8), 672–685. <https://doi.org/10.1016/j.tics.2019.04.013>
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In F. Pereira, C. J. C. Burges, L. Bottou, & K. Q. Weinberger (Eds.), *Proceedings of the 25th international conference on neural information processing systems* (pp. 1097–1105). Curran Associates.
- Krugliak, A., Draschkow, D., Vö, M. L. H., & Clarke, A. (2024). Semantic object processing is modulated by prior scene context. *Language, Cognition and Neuroscience*, 39(8), 962–971. <https://doi.org/10.1080/23273798.2023.2279083>
- LaPointe, M. R. P., Lupianez, J., & Milliken, B. (2013). Context congruency effects in change detection: Opposing effects on detection and identification. *Visual Cognition*, 21(1), 99–122. <https://doi.org/10.1080/13506285.2013.787133>
- MacEvoy, S. P., & Epstein, R. A. (2011). Constructing scenes from objects in human occipitotemporal cortex. *Nature Neuroscience*, 14(10), 1323–1329. <https://doi.org/10.1038/nn.2903>
- Mack, A., Clarke, J., Erol, M., & Bert, J. (2017). Scene incongruity and attention. *Consciousness and Cognition*, 48, 87–103. <https://doi.org/10.1016/j.concog.2016.10.010>
- Mohr, J., Seyfarth, J., Lueschow, A., Weber, J. E., Wichmann, F. A., & Obermayer, K. (2016). BOiS—Berlin object in scene database: Controlled photographic images for visual search experiments with quantified contextual priors. *Frontiers in Psychology*, 7, Article 749. <https://doi.org/10.3389/fpsyg.2016.00749>
- Munneke, J., Brentari, V., & Peelen, M. V. (2013). The influence of scene context on object recognition is independent of attentional focus. *Frontiers in Psychology*, 4, Article 552. <https://doi.org/10.3389/fpsyg.2013.00552>
- Öhlschläger, S., & Vö, M. L. H. (2017). SCEGRAM: An image database for semantic and syntactic inconsistencies in scenes. *Behavior Research Methods*, 49(5), 1780–1791. <https://doi.org/10.3758/s13428-016-0820-3>
- Oliva, A., & Torralba, A. (2007). The role of context in object recognition. *Trends in Cognitive Sciences*, 11(12), 520–527. <https://doi.org/10.1016/j.tics.2007.09.009>
- Ortiz-Tudela, J., Jiménez, L., & Lupiáñez, J. (2023). Scene-object semantic incongruity across stages of processing: From detection to identification and episodic encoding. *Frontiers in Cognition*, 2, Article 1125145. <https://doi.org/10.3389/fcogn.2023.1125145>
- Palmer, T. E. (1975). The effects of contextual scenes on the identification of objects. *Memory & Cognition*, 3(5), 519–526. <https://doi.org/10.3758/BF03197524>
- Peelen, M. V., Berlot, E., & de Lange, F. P. (2024). Predictive processing of scenes and objects. *Nature Reviews Psychology*, 3(1), 13–26. <https://doi.org/10.1038/s44159-023-00254-0>
- Peelen, M. V., & Kastner, S. (2014). Attention in the real world: Toward understanding its neural basis. *Trends in Cognitive Sciences*, 18(5), 242–250. <https://doi.org/10.1016/j.tics.2014.02.004>
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, 51(1), 195–203. <https://doi.org/10.3758/s13428-018-01193-y>
- Press, C., Kok, P., & Yon, D. (2020). The perceptual prediction paradox. *Trends in Cognitive Sciences*, 24(1), 13–24. <https://doi.org/10.1016/j.tics.2019.11.003>
- R Core Team. (2018). *R: A language and environment for statistical computing* [Computer software]. <https://www.rproject.org/>
- Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87. <https://doi.org/10.1038/4580>
- Rensink, R. A., O'regan, J. K., & Clark, J. J. (1997). To see or not to see: The need for attention to perceive changes in scenes. *Psychological Science*, 8(5), 368–373. <https://doi.org/10.1111/j.1467-9280.1997.tb00427.x>
- Rossel, P., Peyrin, C., & Kauffmann, L. (2023). Subjective perception of objects depends on the interaction between the validity of context-based expectations and signal reliability. *Vision Research*, 206, Article 108191. <https://doi.org/10.1016/j.visres.2023.108191>
- Simonsohn, U. (2018). Two lines: A valid alternative to the invalid testing of U-shaped relationships with quadratic regressions. *Advances in Methods and Practices in Psychological Science*, 1(4), 538–555. <https://doi.org/10.1177/2515245918805755>

- Spaak, E., Peelen, M. V., & de Lange, F. P. (2022). Scene context impairs perception of semantically congruent objects. *Psychological Science*, 33(2), 299–313. <https://doi.org/10.1177/09567976211032676>
- Võ, M. L. H. (2021). The meaning and structure of scenes. *Vision Research*, 181, 10–20. <https://doi.org/10.1016/j.visres.2020.11.003>
- Võ, M. L. H., Boettcher, S. E., & Draschkow, D. (2019). Reading scenes: How scene grammar guides attention and aids perception in real-world environments. *Current Opinion in Psychology*, 29, 205–210. <https://doi.org/10.1016/j.copsyc.2019.03.009>
- Võ, M. L. H., & Henderson, J. M. (2009). Does gravity matter? Effects of semantic and syntactic inconsistencies on the allocation of attention during scene perception. *Journal of Vision*, 9(3), Article 24. <https://doi.org/10.1167/9.3.24>
- Võ, M. L. H., & Henderson, J. M. (2011). Object-scene inconsistencies do not capture gaze: Evidence from the flash-preview moving-window paradigm. *Attention, Perception & Psychophysics*, 73(6), 1742–1753. <https://doi.org/10.3758/s13414-011-0150-6>
- Walther, D., & Koch, C. (2006). Modeling attention to salient proto-objects. *Neural Networks*, 19(9), 1395–1407. <https://doi.org/10.1016/j.neunet.2006.10.001>
- Westfall, J. (2015). *PANGEA: Power analysis for general ANOVA designs* [Unpublished manuscript]. University of Texas at Austin. <https://osf.io/x5dc3>
- Wiesmann, S. L., & Võ, M. L. H. (2023). Disentangling diagnostic object properties for human scene categorization. *Scientific Reports*, 13(1), Article 5912. <https://doi.org/10.1038/s41598-023-32385-y>
- Windey, B., Gevers, W., & Cleeremans, A. (2013). Subjective visibility depends on level of processing. *Cognition*, 129(2), 404–409. <https://doi.org/10.1016/j.cognition.2013.07.012>
- Wolfe, J. M., & Horowitz, T. S. (2017). Five factors that guide attention in visual search. *Nature Human Behaviour*, 1(3), Article 0058. <https://doi.org/10.1038/s41562-017-0058>
- Wu, C. C., Wick, F. A., & Pomplun, M. (2014). Guidance of visual attention by semantic information in real-world scenes. *Frontiers in Psychology*, 5, Article 54. <https://doi.org/10.3389/fpsyg.2014.00054>

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